Multimodal Data Management

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Course: Algorithms, Data Structures and Databases

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1. Introduction and Context

The second part of this project builds upon the multimodal music data management pipeline developed in Part 1, extending it toward **embedding generation, evaluation, and analytical interpretation**. While the first phase emphasized data ingestion, normalization, quality control, and architectural design, Part 2 shifts the focus toward **how different data modalities can be represented, compared, and evaluated within a unified analytical framework**.

This phase addresses a central challenge in modern multimedia systems: **how to align heterogeneous representations**, such as textual metadata and audio-derived features, in a way that supports meaningful similarity search and retrieval. Rather than treating embedding models as black boxes, the project adopts an experimental and reflective approach, aiming to understand *how* and *why* different representations behave as they do.

Crucially, the objective of Part 2 is **not to maximize retrieval accuracy**, but to explore embedding strategies, evaluate their behavior using principled metrics, and critically analyze their limitations. This emphasis aligns closely with the educational goals of the course, where understanding system behavior, trade-offs, and failure modes is prioritized over raw performance.

Within this context, the pipeline is extended to support **Text + Audio multimodal embeddings**, enabling controlled experiments that assess cross-modal similarity and alignment in a realistic yet computationally manageable setting.

2. Objectives

The objectives of Part 2 go beyond extending the pipeline with embeddings; they are explicitly oriented toward **understanding multimodal representation behavior** and **evaluating system-level design choices** under realistic constraints. More specifically, this phase aims to:

- **Extend the data pipeline from management to representation learning**, demonstrating how validated and structured data can be transformed into embeddings suitable for similarity-based retrieval.
- **Investigate cross-modal alignment**, focusing on how well semantic information expressed in text aligns with information encoded in numerical audio descriptors once both are embedded into a shared vector space.
- **Design an evaluation methodology grounded in retrieval theory**, using ranking-based metrics that reflect real-world similarity search and recommendation scenarios.
- **Study the impact of representation abstraction**, particularly the consequences of converting audio features into natural-language descriptions.

- **Balance experimental ambition with computational feasibility**, iteratively refining the system to support rapid experimentation.
- **Encourage critical interpretation of results**, treating modest or weak performance as informative outcomes rather than failures.

These objectives intentionally emphasize **analysis and reflection** over optimization. The goal is to expose the complexities of multimodal systems and to reason about their behavior in a principled and transparent manner.

3. Experimental Context and Data Preparation

A core challenge in multimodal systems is deciding **how each modality should be represented** so that meaningful comparison becomes possible. In this project, musical tracks are represented using two complementary but fundamentally different modalities:

1. **Textual metadata**, which encodes semantic, categorical, and contextual information. This modality is inherently symbolic and well aligned with natural-language embedding models.
2. **Audio-related numerical features**, which encode perceptual and acoustic properties but lack direct semantic structure.

Rather than attempting to embed these modalities using separate, specialized encoders, the project adopts a **unification strategy**: audio features are transformed into a textual form that can be processed by the same embedding model as metadata. This choice simplifies integration and enables controlled experimentation on alignment quality.

The conversion from numerical audio features to semantic bins is deliberately coarse. Quantile-based discretization captures relative differences (e.g., low vs. high energy) rather than absolute values, reflecting how humans often reason about music descriptively. Expressing these bins as natural-language phrases further increases interpretability and allows direct inspection of what the embedding model is actually processing.

Data Preparation Considerations

The dataset used for experimentation is intentionally small and curated. By limiting the dataset to 120 tracks with complete metadata and audio features, the project avoids confounding factors such as extreme sparsity or inconsistent coverage. This choice supports **controlled experimentation**, where changes in results can be more confidently attributed to modeling decisions rather than data artifacts.

Additionally, this scale enables repeated experimentation under different configurations, which is essential for understanding model behavior and performance trade-offs.

4. Embedding Generation

Paired Multimodal Embedding Design

The embedding generation process is structured around **explicit pairing**. For each track, two documents are created:

- A **metadata-based text document**, aggregating structured attributes into a single descriptive string.
- An **audio-based semantic document**, composed of natural-language descriptors derived from binned audio features.

Both documents are embedded using the same sentence-level transformer model and indexed together in a vector database. Each pair is linked via a shared identifier, enabling precise evaluation of retrieval behavior.

This design choice is crucial: it defines a **ground truth alignment** at the document level. The system is not asked to infer arbitrary similarity, but rather to retrieve a known paired representation. This makes evaluation unambiguous and well-defined.

Design Trade-offs

Using a single text embedding model for both modalities simplifies the architecture but introduces limitations. The embedding model has strong inductive biases toward linguistic similarity and lacks explicit mechanisms for encoding acoustic relationships. The project embraces this limitation as an experimental variable, allowing its effects to be observed and analyzed.

The embedding process therefore serves not only as a technical implementation step, but also as an **experimental probe** into how representation choices influence retrieval outcomes.

5. Evaluation Methodology

The evaluation framework is designed to reflect how similarity-based systems are actually used in practice. Rather than asking whether a model can assign correct labels, the evaluation asks:

When querying the system with a representation from one modality, how effectively does it retrieve the corresponding representation from another modality?

This framing aligns naturally with **retrieval and recommendation systems**, where ranking quality matters more than absolute classification accuracy.

Metric Interpretation

Each metric used in the evaluation captures a different aspect of system behavior:

- **Recall@K** answers a binary question: *is the correct match visible to the user within the top K results?*
- **Mean Reciprocal Rank (MRR)** captures how early the correct result appears, penalizing late retrievals.
- **Mean and Median Rank** describe the overall distribution of retrieval quality, highlighting whether success is consistent or sporadic.

By using multiple complementary metrics, the evaluation avoids over-reliance on a single performance number and supports more nuanced interpretation.

6. Experimental Results

Experiment 1 – Results Summary

The evaluation of Experiment D produced the following key results:

- Recall@5 and Recall@10 values were low, indicating weak alignment between text and audio representations.
- Mean Reciprocal Rank values remained below 0.05, suggesting that correct audio embeddings were rarely ranked near the top.
- Median rank values were high, often exceeding half of the dataset size.

These results indicate that, while the embeddings encode some shared semantic information, the alignment between modalities is limited.

Experiment 2 – Refined Semantic Audio Descriptors

To improve alignment, a refined version of the experiment (Experiment D2) was conducted. This version enhanced the natural-language descriptors used for audio bins, making them more expressive and context-aware.

Although slight improvements were observed in certain metrics, overall performance remained modest. This outcome highlights the inherent difficulty of aligning structured audio features with textual metadata using a single embedding model.

7. Analysis and Discussion

The experimental results highlight several important interactions between data representation, embedding models, and similarity measures.

First, **same-modality similarity analysis reveals that low-level audio features often produce high similarity scores across different genres**. This indicates that the audio feature space itself does not strongly encode semantic distinctions such as genre or style. As a result, even a perfect alignment between audio features and their textual descriptions would still struggle to recover high-level semantics.

Second, while semantic binning enables compatibility with text-based models, it introduces ambiguity. Phrases such as “high energy” or “medium danceability” are semantically vague and context-dependent. When embedded, these phrases may align weakly or inconsistently with metadata such as genre or artist.

Third, the results emphasize the importance of **model inductive bias**. Sentence-level transformers are trained to capture linguistic meaning, not acoustic structure. When used outside their intended domain, they behave predictably but suboptimally. This observation reinforces the broader lesson that model choice must align with data modality and task requirements.

Taken together, these findings demonstrate that weak alignment is not accidental, but a direct consequence of representation and modeling decisions.

8. Performance and Cost Considerations

Performance considerations played a decisive role in shaping the final design of the experimental pipeline. During the earliest iterations of Part 2, the system exhibited **very long execution times**, in some cases approaching **one hour per run**. These delays significantly hindered experimentation and made systematic evaluation impractical. Several factors contributed to this initial performance bottleneck.

First, the **scale of the Spotify ecosystem** itself imposed practical constraints. Although the experimental dataset was relatively small, enrichment steps relied on querying Spotify-related

metadata and auxiliary resources. Even limited interaction with a platform operating at global scale introduces latency, rate limits, and intermittent access issues that accumulate rapidly across many tracks.

Second, the **download and processing of album cover images** proved to be a major performance bottleneck. Cover images are relatively large binary objects, and retrieving them requires multiple HTTP requests, disk writes, and image processing operations. When executed without strict limits, these operations dominated runtime and made the pipeline unsuitable for rapid iteration.

Third, early pipeline versions attempted to process **all available data without sampling or execution caps**. While conceptually appealing, this approach quickly became infeasible in practice. Embedding generation, indexing, and evaluation costs grow with dataset size, and unrestricted runs made it difficult to test incremental changes or debug specific components.
To address these issues, several deliberate engineering decisions were introduced:

- **Explicit dataset sampling**, ensuring that only a manageable subset of tracks is processed during experimentation.
- **Caps on auxiliary resource downloads**, particularly album covers, which were treated as optional artifacts rather than core analytical inputs.
- **Configurable limits on embedding and evaluation size**, allowing execution time to remain predictable.
- **Environment-variable-driven configuration**, enabling rapid adjustment of performance parameters without code modification.

These changes reduced execution time from nearly an hour to **seconds or a few minutes**, dramatically improving experimental velocity. This evolution highlights an important lesson: in exploratory and experimental systems, **iteration speed is often more valuable than completeness**. Engineering constraints are not merely technical limitations, but active design forces that shape system architecture and experimental feasibility.

9. Limitations

While the system fulfills its experimental and educational objectives, several important limitations and practical challenges must be acknowledged. A central limitation concerns **access to audio information itself**. In early design stages, the project explored the possibility of working directly with raw audio data. However, this approach quickly proved impractical. Accessing raw audio files through commercial platforms such as Spotify is highly restricted, subject to licensing constraints, and often infeasible without specialized agreements or infrastructure. Even when audio previews are available, they are incomplete, inconsistent, and unsuitable for systematic analysis.

As a result, the project transitioned from raw audio processing to the use of **precomputed audio features**. While this shift enabled experimentation, it also introduced a significant abstraction layer. Audio features capture only a limited subset of musical characteristics and omit important aspects such as timbre, harmonic progression, rhythmic structure, and melodic contour. This abstraction directly impacts similarity analysis and cross-modal alignment quality.

Another limitation arises from the **indirect nature of audio–text alignment**. Transforming numerical audio features into semantic textual descriptions enables technical compatibility with text-based embedding models, but it does not guarantee semantic equivalence. The resulting representations are inherently approximate and may fail to capture nuanced musical meaning.
In addition, the system currently lacks a **user-facing interface**. Interaction with the pipeline requires command-line execution and interpretation of logs, metrics, and plots. While appropriate for development and experimentation, this limits accessibility for non-technical users. A graphical or interactive interface would significantly enhance usability, allowing users to explore similarities, inspect embeddings, and understand retrieval behavior intuitively.

Finally, evaluation is limited to **paired retrieval metrics**, which assess alignment quality in a controlled setting but do not directly measure user satisfaction or recommendation usefulness. Extending evaluation to user-centric scenarios would require additional data and interface components. These limitations are not accidental oversights but reflect **deliberate trade-offs** made to balance feasibility, interpretability, and educational value. Acknowledging them explicitly reinforces the project's analytical rigor and clarifies the scope within which conclusions should be interpreted.

10. Conclusions and Next Steps

Part 2 of this project successfully extends the multimodal music pipeline into the domain of **embedding evaluation and experimental analysis**. While alignment results were limited, the project fulfills its educational objective by exposing the complexities of multimodal representation learning and evaluation.
Future work could include:

- Integrating audio-specific embedding models.
- Evaluating additional modalities such as lyrics or user interaction data.
- Expanding evaluation to downstream recommendation tasks.

Overall, this phase demonstrates that building and evaluating multimodal systems is as much about **understanding limitations** as it is about achieving strong performance, reinforcing the project's value as a learning-oriented mini-thesis.